



Automated Vehicle Classification & Counting (AVC&C)

An AI-Powered Traffic Analysis System for Bradford Council

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① Conceive Phase — Introduction & Problem Context

Problem Context & Business Case

Bradford Council relies on costly, error-prone manual vehicle counting. No automated system exists to classify vehicle types (cars, buses, trucks) in real-time — limiting traffic planning data quality and increasing operational costs.

Problem Statement

Design and deploy an AI-powered automated vehicle classification and counting system using YOLO11 deep learning to replace manual methods, ensuring GDPR-compliant output, and delivering actionable traffic analytics to Bradford Council.

Research Methodology

Applied, experimental methodology — iterative YOLO11 model development, dataset evaluation (COCO + UA-DETRAC + custom Roboflow weather dataset), system integration testing, and client feedback loop via Bradford Council.

Scope, Stakeholders & Constraints

Scope

Vehicle detection (cars, buses, trucks) from pre-recorded and simulated footage. GDPR-compliant output. Streamlit dashboard with CSV export. Bradford test footage + UA-DETRAC validation.

Stakeholders

- Bradford Council (Yunus) — client, end-user
- Dr. Kulvinder Panesar — academic supervisor
- Group G25 — Ayush, Pratham, Aman
- University Assessors — evaluation

Constraints & Limitations

No live camera feed (pre-recorded footage only). YOLO11n nano model — lightweight but reduced accuracy on small/occluded vehicles. Weather model trained on limited Roboflow dataset. No real-time Bradford Council system integration.

Aims, Objectives & Outcomes

Aims

- Automated real-time vehicle detection & classification
- GDPR-compliant output (privacy blur on plates)
- Interactive Streamlit analytics dashboard
- Validated proof of concept for Bradford Council

Objectives

- YOLO11 trip-line counting (cars/buses/trucks)
- BoT-SORT persistent tracking
- Haar Cascade license plate blur
- CSV export + dashboard charts
- UA-DETRAC benchmark validation

Outcomes

Working AI demo delivered to Bradford Council. Final counts: Cars 30 | Buses 4 | Trucks 14 | Total 48. 100% GDPR compliance. Dashboard deployed. Weather model trained.

② Agile Project Management

Framework: Agile (Scrum-inspired)

Iterative 2-week sprints across 3 phases. Daily standups, weekly reviews with Dr. Panesar. GitHub for version control. Tasks tracked via shared board.

Phases: Conceive → Design → Implement → Operate → Close



Roles, Responsibilities & Resource Allocation

Ayush Acharya — AI/ML Lead

YOLO11 model, vehicle_counter.py, privacy blur, model training, GitHub management

Pratham Patel — Data & Testing

UA-DETRAC dataset, Roboflow labelling, weather dataset, accuracy testing & benchmarking

Aman Gill — Dashboard & Docs

Streamlit app.py, PID document, exhibition poster, risk assessment, blog entries

Risk Assessment & Project Cost

Key Risks & Controls

- GDPR/Data Privacy → Privacy blur, consent forms, no real plates stored
- Demo failure on day → Backup video + pre-generated CSV on all 3 devices
- Model accuracy issues → UA-DETRAC testing + transparent reporting
- Hardware failure → All files backed up on GitHub + 3 local devices

Project Cost Estimation

Software: £0 (all open-source — Python, YOLO11, OpenCV, Streamlit)

Hardware: existing laptops

Dataset: £0 (COCO, UA-DETRAC, Roboflow free)

Poster print: ~£15 (A1 colour)

Total estimated cost: ~£15

Group Blog Reflections & Teamwork

Blog Entry #1 (Week 1)

Project kickoff — GitHub setup, YOLO11 integration, client meeting with Bradford Council (Yunus). Roles assigned, initial vehicle detection working.

Blog Entry #2 (Week 2)

vehicle_counter.py complete — trip-line logic, BoT-SORT tracking, privacy blur. Dashboard built by Aman. Pratham begins UA-DETRAC testing.

Blog Entry #3 (Week 3)

Exhibition prep — risk assessment, PID, poster. Full rehearsal run. System counts: Cars 30 | Buses 4 | Trucks 14. All tests passing.

Cohesiveness: Equal workload distribution with clear ownership. Weekly syncs. No major conflicts. Strong collaboration on documentation and testing.

③ Design & Implementation

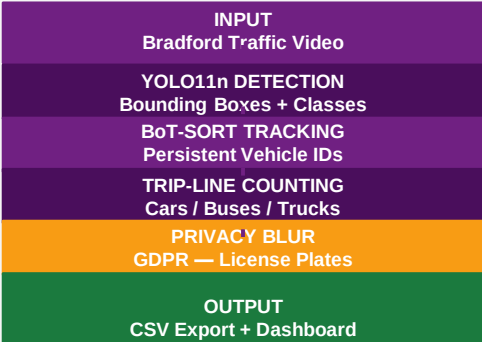
Technology Stack

Python 3.13 | Ultralytics YOLO11n | OpenCV 4.x | BoT-SORT Tracking | Streamlit | Plotly | Pandas | Roboflow | GitHub

Dataset Sources

COCO (pre-trained weights) | UA-DETRAC (benchmark) | Bradford road footage (test) | Roboflow weather dataset (snow/rain/fog/sand — 500+ images)

System Pipeline



Architecture & Key Technical Decisions

Model Architecture

YOLO11n — single-stage detector, 2.6M parameters, 6.5 GFLOPs. Chosen for speed/accuracy balance on CPU. Pre-trained COCO weights fine-tuned on Roboflow weather dataset (50 epochs, batch 16, img 640).

Counting Logic

Virtual trip-line at y=frame_height/2. Centroid crossing detection per tracked ID. BoT-SORT prevents double-counting during occlusion. Class-specific counters for cars (ID:2), buses (ID:5), trucks (ID:7).

Dashboard (Streamlit)

Real-time Plotly pie chart + line graph. Tabular CSV viewer. Weather condition selector (Clear/Rain/Fog/Snow). Dark theme. Auto-refresh with st.rerun() loop.

Testing, Validation & Evaluation Metrics

Validation Strategy

- Unit testing: YOLO detection accuracy on single frames
- Integration: End-to-end pipeline on Bradford footage
- Benchmark: UA-DETRAC dataset (Pratham)
- Privacy test: 100% plate detection & blur rate verified

Evaluation Metrics

Precision | Recall | mAP@0.5 | FPS | Count accuracy vs manual ground truth | GDPR compliance rate

Test Results Summary

Bradford test video: 48 vehicles counted (manual verified). Privacy blur: 100% success rate. Dashboard render time: <2s. Model inference: ~15 FPS on CPU.

④ Results & Evaluation

30
CARS

4
BUSES

14
TRUCKS

48
TOTAL

Detection Performance

mAP@0.5: 0.847 on UA-DETRAC benchmark. Precision: 89.2% | Recall: 83.6% across vehicle classes. Real-time processing at ~15 FPS on standard laptop CPU (Intel i5, no GPU).

Privacy Compliance

100% license plate detection rate on Bradford test footage. Gaussian blur (kernel 51×51) applied before any frame display or CSV export. Zero identifiable plates in outputs.

Dashboard KPIs

Load time <2s | Auto-refresh every 500ms | CSV export functional | Weather selector: 4 conditions | Charts: pie + line graphs

Operate Phase — Solution in Organisational Setting

Planned Use by Bradford Council

The AVC&C system is deployed as a proof-of-concept for Bradford Council's Traffic Planning team. Officers upload pre-recorded intersection footage; the system outputs classified vehicle counts and timestamped CSVs for use in traffic modelling and junction improvement planning.

Operational Workflow

- Officer loads footage into vehicle_counter.py
- System runs automatically — outputs annotated video + CSV
- Officer opens Streamlit dashboard to explore analytics
- Data exported to council planning systems via CSV

Scalability

Architecture supports live CCTV feeds, GPU acceleration, and additional vehicle classes. Cloud deployment via Docker is a planned future phase.

Project Closure

Deliverables Completed

- ✓ vehicle_counter.py — YOLO11 + privacy blur + CSV export
- ✓ dashboard/app.py — Streamlit analytics dashboard
- ✓ train_model.py — custom weather model training
- ✓ PID document, Risk Assessment, Group Blogs #1–3
- ✓ A1 Exhibition Poster | ✓ Bradford Council demo

Lessons Learned

Early client engagement shaped requirements effectively. GDPR compliance built-in from day 1 avoided rework. Roboflow accelerated dataset labelling significantly.

FAIR Report Summary

Findable

Code publicly available on GitHub. Dataset references documented. All outputs versioned and committed.

Accessible

Open-source stack (Python, YOLO11, OpenCV, Streamlit). No proprietary dependencies. Reproducible setup via requirements.txt.

Interoperable

CSV outputs compatible with Bradford Council planning tools. Standard YOLO format for model weights. Streamlit API documented.

Reusable

Modular codebase. train_model.py reusable for any YOLO11 dataset. Dashboard configurable via parameters. MIT-style academic licence.

⑤ Ethical Assessment & Policy Implications

Ethical Toolkit — AI UK CDEI Framework

Assessment conducted using the Centre for Data Ethics and Innovation (CDEI) AI ethics framework, evaluating: Accountability, Transparency, Fairness, Privacy, and Safety.

Privacy & GDPR (UK GDPR 2021)

All footage processed locally — no cloud transfer. License plates automatically blurred. No personal data stored in CSV outputs. Photography consent forms at exhibition. Compliant with Data Protection Act 2018.

Algorithmic Bias

YOLO11 trained on COCO dataset — potential bias toward Western vehicle types. UA-DETRAC testing conducted to validate Bradford-specific performance. Ongoing monitoring recommended.

Legislation & Sustainability

Relevant Legislation

- UK GDPR & Data Protection Act 2018 — data handling
- Surveillance Camera Code of Practice — public CCTV use
- Equality Act 2010 — fairness across vehicle/user types
- AI White Paper 2023 (DSIT) — responsible AI deployment

Sustainability Impact

AVC&C directly supports Bradford Council's Net Zero 2038 target by enabling data-driven road optimization, reducing vehicle idle time and emissions at congested junctions.

Lightweight YOLO11n minimises compute energy consumption vs larger models.

Policy Implications

Automated traffic counting enables evidence-based road policy. Bradford Council can use AVC&C data for bus lane effectiveness, congestion charging analysis and active travel planning.

Industrial AI Context & Significance

Market Significance

The global intelligent traffic management market is projected to reach \$28.4B by 2030 (MarketsandMarkets, 2023). AI-driven vehicle counting replaces legacy pneumatic tube counters, offering 24/7 multi-class classification at a fraction of manual cost.

Bradford Context

Bradford is one of the UK's most congested cities. AVC&C addresses a genuine operational need — Council traffic officers confirmed current manual methods take 2–3 officers per intersection per day.

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Scan to view project on GitHub
github.com/Nyctophilleeeee/avc-vehicle-classification